Problem Set 10: Sampling Distributions and Project Setup

SPEA V706 Math Camp | Module 10 · Friday, August 14

The last set. **Part A** is the algebra of estimators. **Part B** has you build a real project from an empty folder and then use it to watch the CLT happen in your own simulation.

Reading: Hansen, *Probability and Statistics for Economists*, chapters 6–7. Decks: **M5 Sampling Distributions**, **W5 Project Setup**.

Part A. Concepts

A1. The sampling distribution of a mean

A population has mean $\mu = 20$ and standard deviation $\sigma = 6$. You draw an iid sample of size $n = 36$ and compute $\bar{Y}_n$.

a. What is $E[\bar{Y}_n]$? Which property is this?
b. What is $\text{Var}(\bar{Y}_n)$?
c. What is the standard error of $\bar{Y}_n$?
d. How large would n have to be to get the standard error down to 0.5?
e. Is $\bar{Y}_n$ Normally distributed here? Say exactly what you can and cannot conclude from the information given.

A2. The rate

a. Starting from $\text{se}(\bar{Y}_n) = \sigma/\sqrt{n}$, show algebraically that halving the standard error requires quadrupling n.
b. A colleague has 1,000 observations and a standard error of 0.40. They want 0.10. How many observations do they need in total? How many more than they have?
c. One sentence on what this implies for the returns to collecting more data.

A3. True or false

For each statement, say true or false and give one sentence of justification.

a. The law of large numbers says that if a fair coin comes up heads ten times in a row, tails is now more likely on the next flip.

b. The central limit theorem says that with a large enough sample, the variable Y itself becomes Normally distributed.
c. The central limit theorem requires the population distribution to be Normal.
d. If an estimator is unbiased, then the estimate from my particular sample is close to the truth.
e. As n grows, $\text{Var}(\bar{Y}_n)$ goes to zero.
f. The standard deviation of Y and the standard error of $\bar{Y}_n$ measure the same thing in different units.

A4. Unbiasedness, line by line

Show that $E[\bar{Y}_n] = \mu$ when $Y_1, \ldots, Y_n$ are iid with mean μ. Write out each step and name the expectation rule (E1, E2, or E3) that justifies it. Then answer: which part of the iid assumption did you use, and which part did you not?

A5. When the approximation is poor

The CLT holds in the limit, and "large enough n" depends on the population.

a. Give two features of a population distribution that would make $n = 30$ too small for the Normal approximation to be any good.
b. Give one feature of the *sampling scheme*, rather than the population, that can break the σ^2/n formula entirely.
c. In the `wage1` data, the standard deviation of `wage` is about \$3.69 and there are 526 workers. Compute the standard error of the mean wage. Then say in one sentence what each of those two numbers describes.

Part B. R

This part is graded on the folder, not just the answers. Build it properly.

B1. Build a project from scratch

Create a new folder called `sampling_lab` and set it up the way deck **W5** describes.

a. Create the RStudio project file (`sampling_lab.Rproj`) with File > New Project > New Directory.
b. Create subfolders: `code/`, `outputs/figures/`, and `raw_data/`.
c. Create a `.gitignore` that excludes `raw_data/` and `.Rproj.user/`.
d. Create an empty `main.r` at the top level.
e. Explain in one sentence why `here::here("code", "01_simulate.R")` is better than `setwd("/Users/yourname/Desktop/sampling_lab")`.

B2. `code/01_simulate.R`

Write a script that draws sampling distributions of the mean.

The population is Exponential with rate 1, so $\mu = 1$ and $\sigma = 1$. For each of $n = 5$, $n = 30$, and $n = 200$, draw 5,000 samples of that size and record the mean of each.

```r
# code/01_simulate.R
pacman::p_load(data.table, here)
set.seed(20260814)

n_sims <- 5000
sizes  <- c(5, 30, 200)

# your code: for each size, 5000 sample means
# result: a data.table with columns n and xbar (15000 rows)

saveRDS(sims, here("outputs", "sampling_sims.rds"))
message("01_simulate.R complete")
```

a. Fill in the simulation. `replicate()` or `matrix()` plus `rowMeans()` both work; the matrix version is much faster.
b. Confirm your result has 15,000 rows and the columns you expect.

B3. `code/02_analyze.R`

a. For each n, compute the mean of `xbar` across the 5,000 samples. Compare each to the true population mean of 1. What does the LLN predict?
b. For each n, compute the standard deviation of `xbar`. Compare each to the theoretical $\sigma/\sqrt{n} = 1/\sqrt{n}$. Report the two side by side in a small table.
c. Check the rate directly: divide the standard deviation at $n = 5$ by the one at $n = 200$. The sample sizes differ by a factor of 40, so what ratio should you see? Does the simulation agree?
d. Make one figure showing all three sampling distributions, save it to `outputs/figures/` with `ggsave()`.

B4. Tie it together

a. Write `main.r` so it sources both scripts in order using `here()`.
b. Restart R (Session > Restart R), then run `main.r` from a clean session. Confirm it produces the saved figure with no errors and no objects left over from your interactive session.
c. Delete `outputs/sampling_sims.rds` and the figure, then run `main.r` again. Everything should come back. If it does not, something in your pipeline depends on state you did not write down.

B5. Read your own figure

a. At which n does the sampling distribution still look visibly skewed?
b. At which n does it look Normal?
c. The population you sampled from is strongly right-skewed and nothing like a bell curve. Explain in two or three sentences why the sample means look Normal anyway, and what would have to be true about the population for this to fail.

Check yourself. Without notes:

1. What is the difference between a standard deviation and a standard error?
2. Why does the sample mean have variance σ^2/n rather than σ^2?
3. What does the CLT let you do that the LLN alone does not?